Cube-Sum Permutations: An 82-Identity Elementary Certificate

Qichao Wang

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Abstract

Let $P(n)$ count permutations p of $\{1,\dots,n\}$ for which every $i+p(i)$ is a perfect cube. We prove $P(n)>0$ for every $n\ge 173$ from exactly 82 explicit endpoint-multiset identities, comprising 1,030 reflected integer intervals, plus an elementary suffix-reflection rule and a cubic-gap inequality. An explicit predecessor table explains all remaining finite lengths without a matching search, a large permutation archive, or dependent augmenting paths. The finite case distinctions are *not* eliminated: this is a solver-free, hand-checkable certificate rather than a case-free parameterized construction.

1 Two elementary lemmas

Lemma 1 (Boundary cancellation)**.** *Let $0=b_0<b_1<\cdots<b_t=n$ and let $c_1,\dots,c_t$ be positive integer cubes. Set $L_j=c_j-b_j$ and $R_j=c_j-b_{j-1}$. Assume $1\le L_j<R_j\le n+1$. Then $p(i)=c_j-i$ for $b_{j-1}<i\le b_j$ is a cube-sum permutation if and only if*

$$\boxed{\{1,R_1,\dots,R_t\}_{\text{multi}}=\{n+1,L_1,\dots,L_t\}_{\text{multi}}.}$$

Proof. Each input block maps bijectively onto the integer interval $[L_j,R_j)$. At an integer x, the jump in the total interval-coverage count is the multiplicity of x as a left endpoint minus its multiplicity as a right endpoint. The displayed identity makes that jump 1 at $x=1$, -1 at $x=n+1$, and zero everywhere else. Thus coverage is exactly one at every $x\in[1,n]$. Reversing this argument proves necessity. □

Lemma 2 (Growing cubic suffix)**.** *Suppose $0\le m<n$, $m+n+1=c$ is a cube, and a cube-sum permutation on $[1,m]$ exists (the $m=0$ permutation is empty). Keep the existing permutation on $[1,m]$ and set $p(i)=c-i$ for $m<i\le n$. Then p is a cube-sum permutation on $[1,n]$.*

Proof. The new reflection is a bijection from $[m+1,n]$ to itself, and each new sum is c. □

2 An explicit 82-identity finite certificate

The appendix records 82 rows of the form $n\ b_1:k_1\ \cdots\ b_t:k_t$. The corresponding cubical sums are $c_j=k_j^3$ and the first boundary is $b_0=0$. For every row, the boundaries increase, $1\le L_j<R_j\le n+1$, and the multiset equality in Lemma 1 holds exactly. These are 1,030 simple integer-interval identities. Exactly 22 listed rows have $n<173$ and exactly 60 have $173\le n\le 364$.

The following two lists give every omitted base length without further matching data. For any tabulated predecessor m under cube c, define $n=c-m-1$ and apply Lemma 2. The lists are ordered by cube, so every predecessor is either a seed or was obtained in an earlier row.

First: 26 reflected lengths below 173.

cube	already constructed predecessors m
8	0
27	0,7
64	0,7,19,26
125	0,7,19,26,37,44,56
216	44,56,63,66,68,80,82,85,87, 100,103,105

Together with the 22 lower seeds, these supply 48 distinct lengths below 173.

Second: 49 reflected lengths in $[173, 255] \cup [339, 364]$.

```
cube     previously constructed predecessors m
216      0,7,19,26,37
343      0,87,98,100,103,105,110,112,
         115,116,117,118,119,120,121,122,
         123,124,125,126,128,129,130,131,
         132,133,135,147,149,150,151,152,
         155,156,159
512      147,149,150,151,152,155,156,
         159,171
```

The 60 upper seed rows and these 49 reflections are exactly the 109 integers in $[173, 255] \cup [339, 364]$. There is no circular dependency: each predecessor m is strictly smaller than its constructed length n.

For example, $211 = 343 - 131 - 1$, $340 = 512 - 171 - 1$, and $342 = 343 - 0 - 1$. The first example replaces a previously necessary-looking 53-segment independent row by one suffix reflection.

Lemma 3 (Complete finite interval)**.** *For every integer $173 \le n \le 364$, a cube-sum permutation of $[1, n]$ exists.*

Proof. The explicit arithmetic rows and the two predecessor tables establish the claim for $[173, 255] \cup [339, 364]$. For each of the 83 remaining lengths $256 \le n \le 338$, take $m = 511 - n \in [173, 255]$. Then $m < n$ and $m + n + 1 = 512 = 8^3$, so Lemma 2 applies. □

3 A purely symbolic infinite tail

Theorem 1. *For every integer $n \ge 173$, there is a permutation p of $[1, n]$ such that $i + p(i)$ is a perfect cube for all indices i.*

Proof. The base interval is Lemma 3. For $n \ge 365$, assume the claim for every smaller integer at least 173, and let k^3 be the least integer cube satisfying $k^3 \ge n + 174$. Thus $m = k^3 - n - 1 \ge 173$.

If $365 \le n \le 555$, then $k = 9$ and $k^3 = 729 \le 2n$. For $n \ge 556$, minimality of k implies $k \ge 10$ and $n \ge (k-1)^3 - 173$. Therefore

$$\begin{aligned} 2n - k^3 &\ge 2(k-1)^3 - 346 - k^3 \\ &= k^3 - 6k^2 + 6k - 348 \\ &\ge 112 > 0. \end{aligned}$$

The final polynomial equals 112 at $k = 10$ and increases thereafter: its first difference is $3k^2 - 9k + 1 > 0$. In either case $k^3 \le 2n$ and hence $m < n$. Apply the induction hypothesis at m and Lemma 2. □

Logical scope. The theorem uses explicit finite arithmetic equalities and short symbolic lemmas. The optional script `verify_82_seed_proof.py` checks the 82 rows and finite closure without matching or other external data. It is not a premise for the proof. A uniform, case-free formula replacing the 82 rows is *not* claimed.

A All 82 explicit integer-multiset witnesses

Every token $b : k$ means $p(i) = k^3 - i$ up to endpoint b. For each row, verify the same endpoint balance of Lemma 1.

66: 2:4 5:2 58:4 61:5 63:4 66:5

82: 2:4 5:2 21:3 42:4 61:5 63:4 82:5

85: 2:4 24:3 39:4 61:5 63:4 85:5

100: 2:4 24:3 61:5 63:4 100:5

103: 2:4 5:2 21:3 61:5 63:4 103:5

116: 2:4 3:3 4:2 8:3 9:5 10:3 11:5 15:3 16:5 17:3 18:5 22:3 23:5 24:3 61:5 63:4 100:5 101:6 102:5 106:6 107:5 108:6 109:5 113:6 114:5 115:6 116:5

118: 1:3 2:2 3:3 4:2 5:3 6:2 7:5 8:3 9:5 10:3 11:5 12:3 14:5 15:3 16:5 17:3 18:5 19:3 21:5 22:3 23:5 24:3 25:5 26:3 98:5 99:6 100:5 101:6 102:5 103:6 105:5 106:6 107:5 108:6 109:5 110:6 112:5 113:6 114:5 115:6 116:5 117:6 118:5

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120: 2:4 3:3 4:2 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 120:5

121: 2:4 3:3 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 121:5

122: 2:4 61:5 63:4 122:5

123: 2:4 31:5 32:4 61:5 63:4 92:5 93:6 122:5 123:6

125: 2:4 3:3 4:2 9:5 10:3 16:5 17:3 23:5 24:3 29:5 34:4 61:5 63:4 90:5 95:6 100:5 101:6 107:5 108:6 114:5 115:6 120:5 125:6

126: 2:4 5:2 28:5 35:4 61:5 63:4 89:5 96:6 119:5 126:6

127: 1:5 4:2 5:3 6:2 11:5 12:3 18:5 19:3 25:5 26:3 27:5 30:4 31:5 36:4 88:5 91:6 92:5 97:6 102:5 103:6 109:5 110:6 116:5 117:6 118:5 121:6 122:5 127:6

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150: 3:5 4:2 30:4 33:5 59:4 65:5 91:6 94:5 121:6 124:5 150:6

151: 3:5 30:4 33:5 60:4 64:5 91:6 94:5 121:6 124:5 151:6

155: 3:5 30:4 33:5 60:4 91:6 94:5 121:6 124:5 155:6

156: 3:5 4:2 30:4 33:5 59:4 91:6 94:5 121:6 124:5 156:6

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188: 2:5 5:4 7:2 30:4 32:6 61:4 91:6 93:5 154:6 157:7 159:6 188:7

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